

Phase 2

Design

Tuesday Group 7

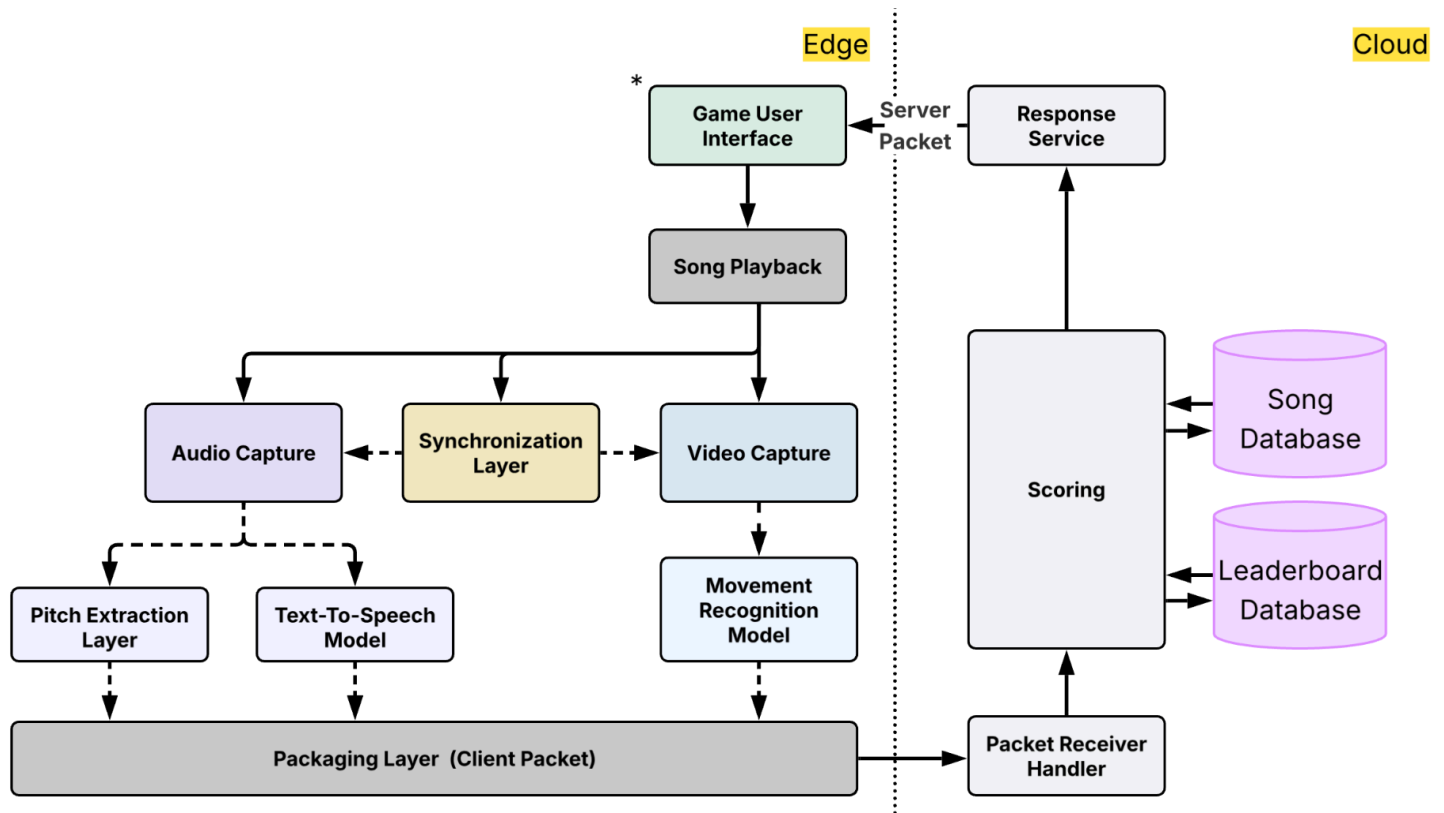
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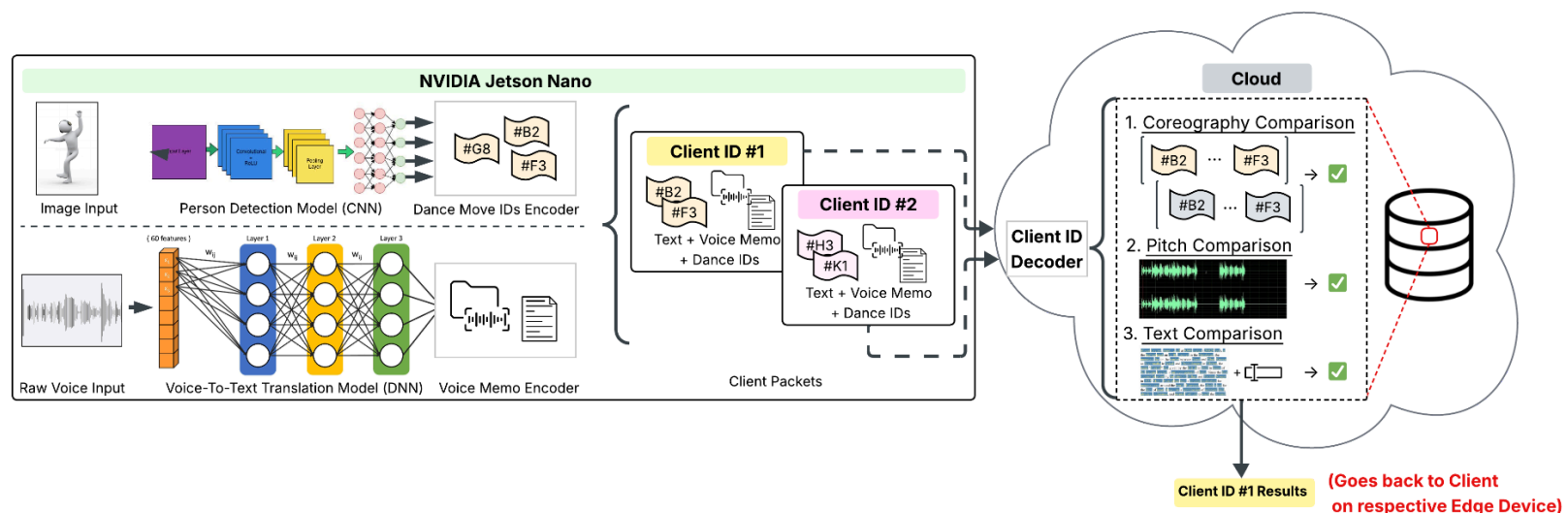
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Device System Design Diagram



Workload Distribution Diagram



EDGE Workload

Input	Computation	Output
- Record player movements via camera sensor - Record player audio via audio sensor - Server packets consisting of: - Final player score - Text accuracy - Pitch accuracy - Choreography accuracy - Score relative to online leaderboard	- Takes recorded player movement and transforms it into a person segmentation frame using a CNN. - Takes recorded player audio and extracts the text and pitch waveforms using a DNN. - Encodes player segmentation frames, speech-to-text data, and vocal pitch information. - Extracts the accuracy of player segmentation frames, speech-to-text processing, and pitch detection.	- Displays dance choreography along with the song's lyrics. - Client ID packets consisting of: - Unique client ID's - Player segmentation frames data - Encoded speech-to-text data - Encoded pitch data

CLOUD Workload

Input	Computation	Output
- Client ID packets consisting of: - Unique client ID's - Player segmentation frames data - Encoded speech-to-text data - Encoded pitch data	- Extracts player segmentation frame, encoded speech-to-text data, and encoded pitch data, and labels them according to the corresponding unique client ID. - Decodes speech-to-text data and pitch data. - Computes the accuracy of player segmentation frames compared to the song choreography segmentation sets. - Computes the accuracy of player speech-to-text and pitch compared to the song lyrics and pitch. - Updates global leaderboard	- Server packets consisting of: - Final player score - Text accuracy - Pitch accuracy - Choreography accuracy - Score relative to online leaderboard

Technical Worksheet (Tool Inventory)

Edge Layer:

- Jetson Nano: main edge unit used for audio/visual computations from its inputs
- Microphone (audio sensor): Constantly captures user's voice, used as input for DNN model for voice to text
- WebCam (camera sensor): Collect videoframes of user's real-time movements, used as inputs for CNN model for model creation
- CNN (Person Detection Model): ML model that transforms video frames into player segmentation models
- DNN (Voice-To-Text Translation Model): ML model that translates audio files into text, along with extracting pitch waveforms
- Client Packets: Computed information from both models, unique client ID's, player segmentation frame data, encoded speech-to-text data, and encoded pitch data all packaged together, ready to be shipped to the cloud layer.

Cloud Layer:

- Cloud Server: Used to unpack client packets for evaluating user's game inputs
- Leaderboard Database: Stores high scores of each song associated with the respective user.
- Song Database Stores song mp3s, choreography, and lyrics.
- Server Packets: Encoded data that consists of Final player score, text accuracy, pitch accuracy, choreography accuracy, and score relative to online leaderboard sent to the edge device for outputting relevant information to the UI.

Scrum Sprint Report

Sprint Goal:

The goal for this sprint was to move from our initial project proposal to a detailed system design and project workflow. During this sprint, we established weekly meeting times, defined the overall workflow of the project, and assigned individual responsibilities to each team member to ensure the work was distributed evenly and that development could proceed in an organized manner.

Summary:

So far, as a team, we have met to set up weekly meeting schedules, discuss the workflow of the project, and design the overall system architecture. We are breaking down tasks for each team member and assigning an even amount of work:

- Edward is in charge of the audio recognition model.
- Z is in charge of the human recognition model.
- Andy is in charge of the server.
- Hanson is in charge of the UI.

This workload distribution covers approximately 90 percent of the project. Once we complete these individual components, we will stitch them together to form the complete system. If a group member finishes their assignment early, they are expected to help another group member if needed, or begin working on a new task. So far, we have not started coding because we were wrapping up the lab exercises for this class. However, over the weekend and by Phase 3, we plan to have much of the code completed.

Phase 2 Contributions:

- Edward worked on the Device System Design Diagram,
- Z worked on the Workload Distribution Diagram
- Andy worked on the Technical Worksheet (Tool Inventory)
- Hanson worked on the Scrum Sprint Report

With that being said, we all contributed equally to the work leading up to the Phase 2 documentation and diagrams. We worked together to decide on variable naming conventions and the overall flow of the project and system to ensure that we are all on the same page before beginning implementation.